

SYSTEMS ENGINEERING

Systems Analysis & Design

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Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



Outline

- 1 Basic Concepts
- 2 Human Activities •
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



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- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
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What is Systems Engineering?

- **Systems Engineering** is a discipline that studies the design, implementation, and maintenance of complex systems.
- This discipline is based on interdisciplinary fields, such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
- **Systems Engineering** is a holistic approach to engineering that focuses on how to design and manage complex systems over their lifecycle.

design



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What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields, such as **control engineering**, **industrial engineering**, **software engineering**, **mechanical engineering**, **electrical engineering**, **organizational studies**, **project management**, and **others**.
- **Systems Engineering** is a **holistic** approach to **engineering** that focuses on how to **design** and manage **complex systems** over their **lifecycle**.



Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for designing, implementing, and maintaining complex systems.
- A **Systems Engineer** must have a broad understanding of engineering, mathematics, science, and technology.
- A **Systems Engineer** must be able to analyze and solve problems in a systematic and holistic way.

architect
tech lead
pre-sales

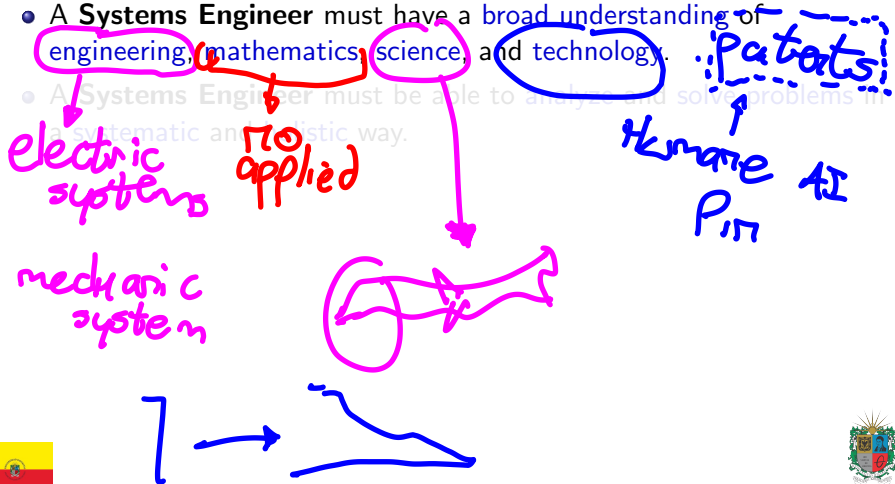
software
engineering

Academic & researchers



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Programming
Contest

Saturday
12 - 18

10 Problems

English

1 kb

100

Google

Apostol



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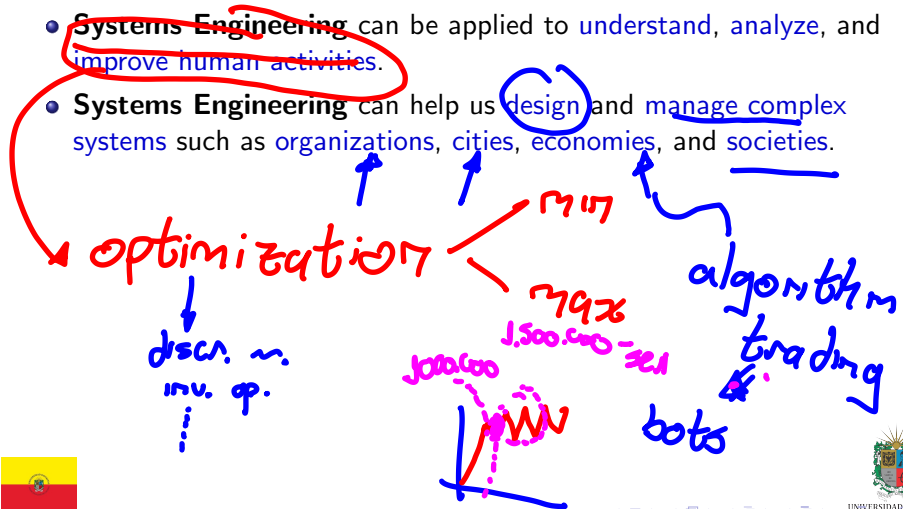
Human Activities

- **Human activities** are **complex systems** that involve **multiple components, interactions, and feedback loops**.
- **Systems Engineering** can be applied to understand, analyze, and improve human activities.
- **Systems Engineering** can help us design and manage complex systems such as organizations, cities, economies, and societies.



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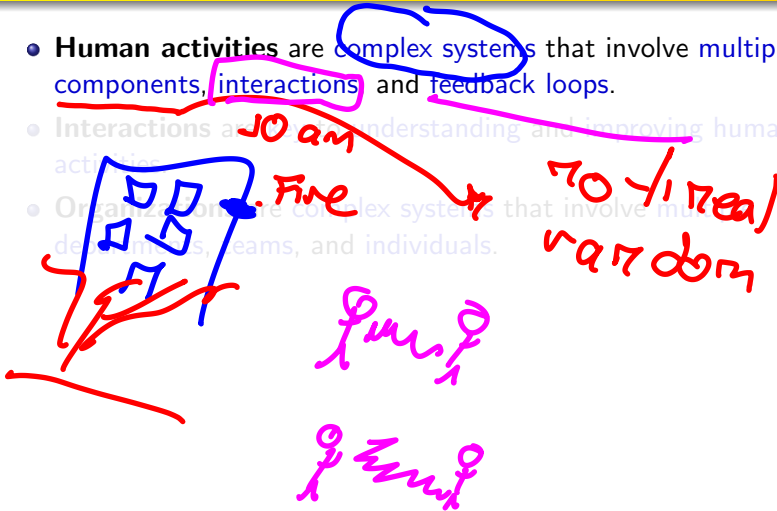


Human Activities: Complexity

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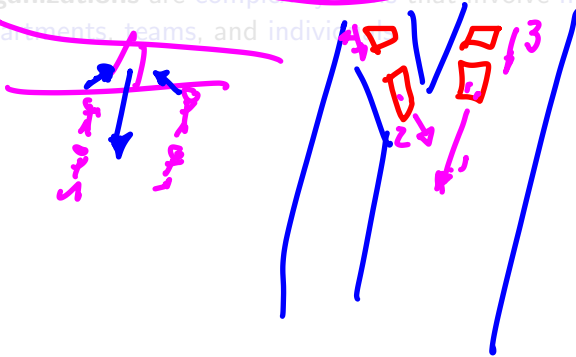
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- Organizations are complex systems that involve multiple departments, teams, and individuals.



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different points of view
 different dynamics
 not all days are the same



Organizations as Systems

- **Organizations** can be **viewed** as: a rational system, a natural system, or an open system.
 - environment
 - logic
 - human
- A rational system is a formal organization that is designed to achieve specific goals as a machine.
- A natural system is an informal organization that is emergent and adaptive based on human interactions.
- An open system is an organization that is interconnected with its environment and adapts to changes.



Organizations as Systems

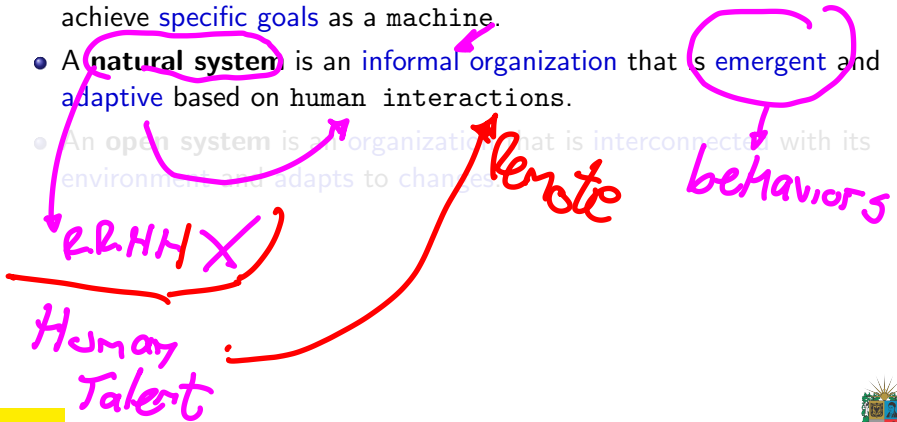
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mission
vision
goals



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clients
suppliers
competence
↳ rivals

change
plan

startup



Synergy

- **Synergy** is a simple but powerful concept: the whole is greater than the sum of its parts.
- It means the **interactions** could boost the capabilities of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define improvements in **systems**.

6-degree separation

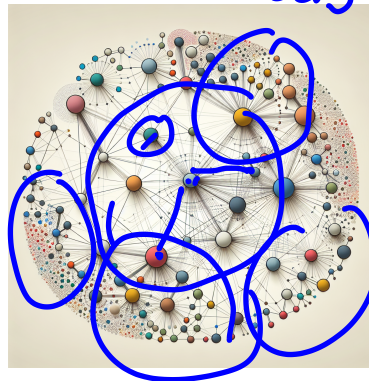


Figure: Prompt: Define a draw of clusters in social networks.



Synergy

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- It means the **interactions** could **boost the capabilities** of the parts of the **system**. Also, it lets both understand **emergent behaviors** and define **improvements** in **systems**.
- One of the main concepts is the **theory of computation**. Based on **graphs**, you could define a **computational machine**.

*Algo
Turing*

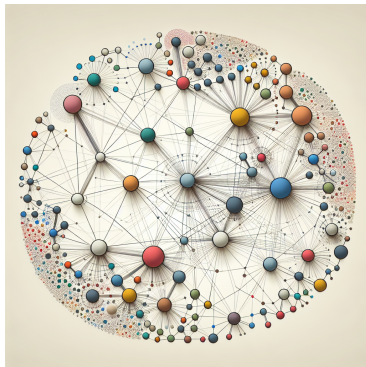
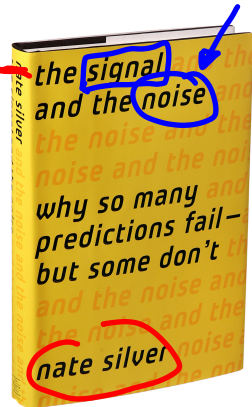
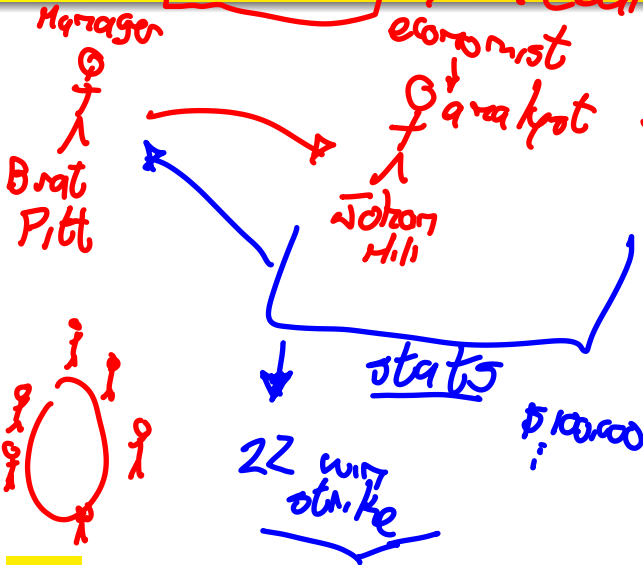


Figure: Prompt: Define a draw of clusters in social networks.



Synergy: Money Ball → Baseball

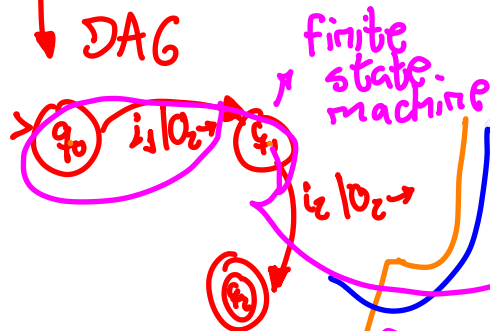


Talking with Machines!



Algorithm

↓
DAG



Alonzo Church

- **Alan Turing** proposed a hundred years ago an **Universal Machine**, capable of take any algorithm defined as a state machine, and process it in a binary language. → **Digital**

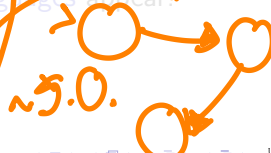
Forty years after, Noam Chomsky proposed the formal languages based on generative grammars. Here is where the high-level programming languages appear.

01010110110

algo

0101010
input

parameter



Talking with Machines!

$\Sigma = \{a, b\}$

$G = \begin{cases} S \rightarrow aV_1 \\ V_1 \rightarrow ba|ab|V_2 \\ V_2 \rightarrow \vdots \end{cases}$

all words of
a language

tree

graph
(FSM)

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→ binary → UTM



Programming Languages

- **Programming Languages** with more capabilities, easier comprehension had been created. Also, more people start to code into specific domain programming languages.

Natural Language

Java: hungry? lunch: home;
Python:



lunch if hungry else home

Demo time!

General-purpose

CD (Raggs)

SQL

Terraform

(IaC)



Programming Languages

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Demo time!

- **Andrej Karpathy**, a leading figure at **tesla** and now at **OpenAI**, said: *Nowadays, English is the most important programming language.*

NLP

LLM code

agent intelligent

vibing code

A sand B GPS C reported D fish



Applications of Systems Engineering

- **Systems Engineering** can be applied to understand, analyze, and improve complex systems.
- **Systems Engineering** can be applied to design and manage complex systems such as organizations, cities, economies, and societies.
- **Systems Engineering** can be applied to design and manage complex systems such as software, hardware, and networks.

cloud
IoT
Robotics

Ubiquitous → IoT
↳ Woatras
> 93 Countries
> 7 Petabytes/week



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Cibernetics

~1448

- **Cibernetics** is the study of systems, control, and communication in animals, machines, and organizations.

- Cibernetics is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.

- Cibernetics is the foundation of systems engineering and information technology.

environment

auto-regulation

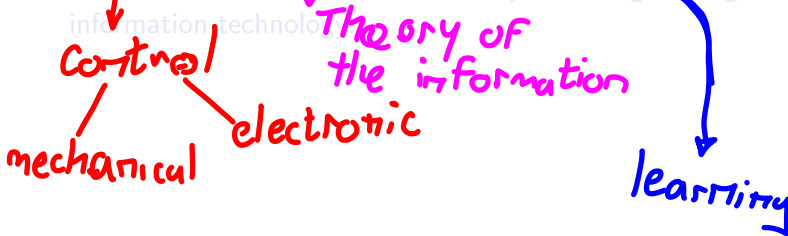


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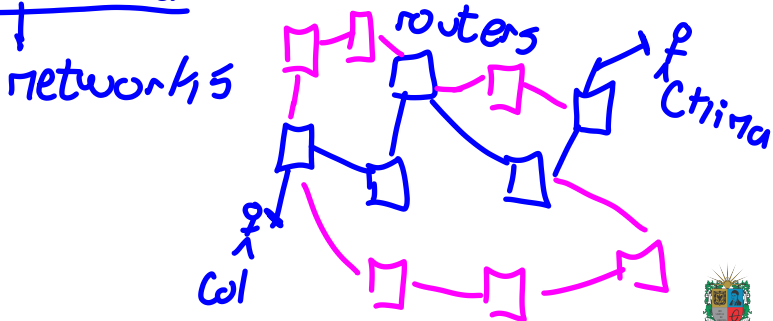
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Technology

- **Technology** is the application of scientific knowledge to solve problems and improve systems.

Temporal is the key to dealing with complex systems
such as organizations, cities, economies, and societies.

~ 2000-3000 years ago



100
WWI



Technology

- **Technology** is the *application* of scientific knowledge to solve problems and improve systems.
- **Technology** is the key to designing and managing complex systems such as organizations, cities, economies, and societies.

Computers as
tools

Comp.
science

math
phys.
bido.
Chem.



Technology: **AI**

- **Artificial Intelligence** is a *field* of computer science that studies how to design and implement intelligent agents.

Artificial Intelligence is the
 solve problems
 like a human ~

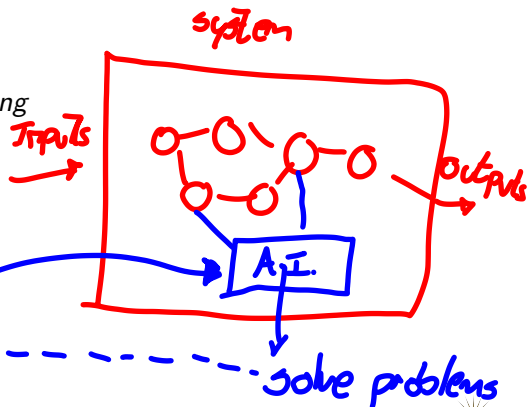
software
 program
 who solve problems
 → reasoning
 → intuition/reflex

→ maths
 ↳ statistical
 ↳ probabilistics
 ↳ discrete
 ↳ non-linear
 → biology
 ↳ arm
 ↳ eye-lens
 → psychology
 ↳ learning

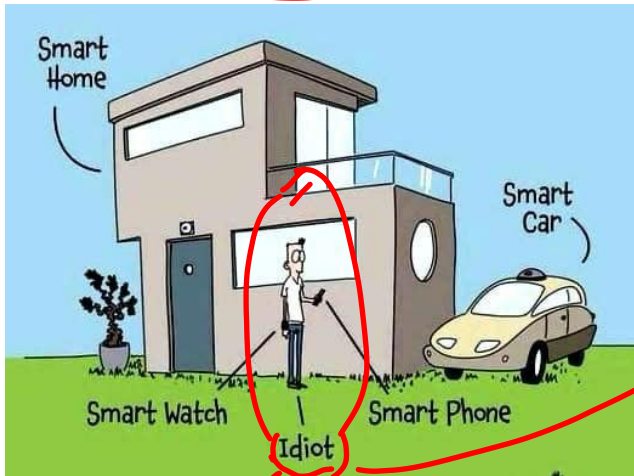


Technology: AI

- **Artificial Intelligence** is a *field* of computer science that *studies* how to *design* and *implement* intelligent agents.
- **Artificial Intelligence** is the key to *designing* and *managing* *complex systems* such as smart-organizations, smart-cities, smart-economies, and smart-societies.



Artificial Intelligence **not** as a System



Chat
GPT
↓
less
reasoning
capacity

Wall-e



Cibernetics & Technology in Systems Context

- **Cibernetics** and **technology** are the foundation of systems engineering and information technology.
- **Cibernetics** and **technology** are the foundation of artificial intelligence and smart systems.

//
sciences
control
rage



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Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to make decisions and solve problems.
- **Team-based structure organizations** are flexible, agile, and innovative because they **empower employees** and **encourage collaboration**.



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Teams as a System

- **Teams** are a **system** where collaboration (synergy) and communication are **key** to **success**.
- Defining and automating the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.



Soft Skills

- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and harmoniously with other **people**.
- Typical Soft Skills:
 - **Communication** skills (verbal and written).
 - **Teamwork** and collaboration.
 - **Problem-solving** and critical thinking.
 - **Adaptability** and flexibility.
 - **Time management** and organization.
 - **Leadership** and management.
 - **Emotional intelligence**.
 - **Creativity** and innovation.
 - **Conflict resolution**.
 - **Networking** and relationship building.
 - **Customer service** and client management.



Computer Analyst

● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.



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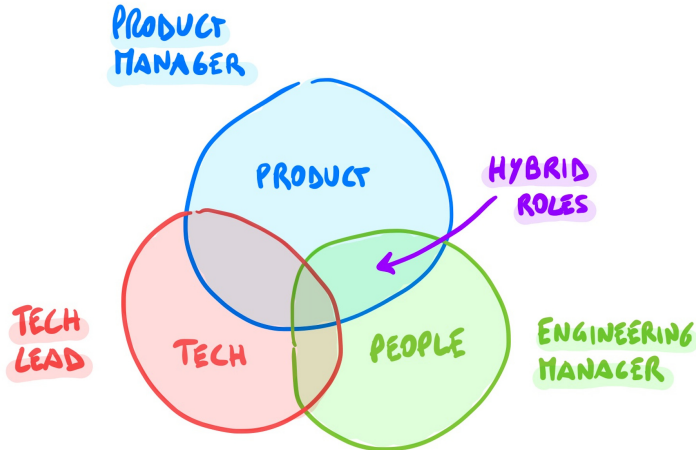
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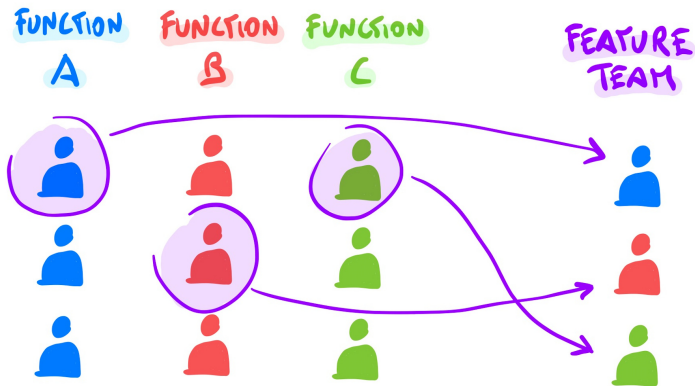
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Tech Company Typical Structure



Feature Teams



What is to be a leader? I

- **Leading** a team is **not a role**. It is a decision, you could be a **leader** anytime and anywhere.
- **Teamwork culture** is pretty **important**. It creates habits, open communication, safety spaces for inclusion.
- **Psychological safety** is a **key** point to have an **effective** team. You could develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. Hierarchy exists by **status** and **power**.
- In a **hierarchy experts lead** to make **better decisions**. However, anyone must be **careful** to not **leave people behind**.



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- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just complete tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish — just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for self-growth are vital.
- Making **ethical decisions** is **key**; it leads to taking the **right** and **better actions**.



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What is to be a leader? III

- It is important to maintain **psychological well-being**. You will be **stronger**, better able to **help** people, and have a better **perspective** on everything.
- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



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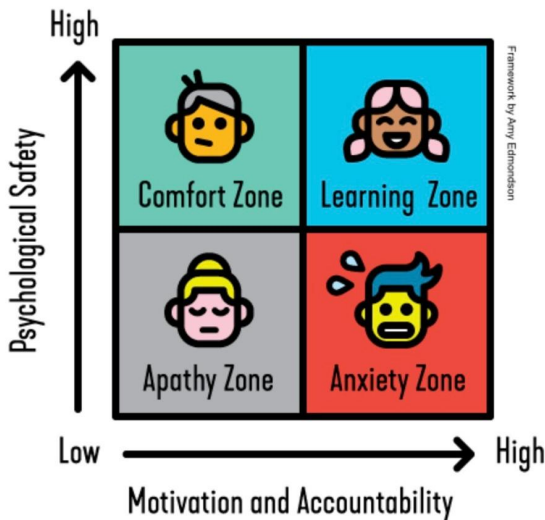


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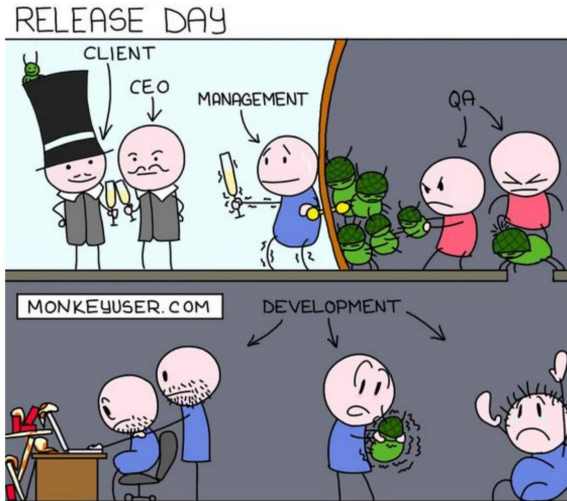
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Working Zones



Real World!



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

